

INÉS GARCÍA REDONDO

ines.garciaredondo@unifr.ch • Personal Webpage

ACADEMIC POSITIONS

Senior researcher (100%)

University of Fribourg, Department of Informatics

November 2025 - October 2027

EDUCATION

Imperial College London

PhD in Mathematics fully funded by the London School of Geometry and Number Theory

2021–2025

Complutense University of Madrid

MSc in Advanced Mathematics

2020–2021

Average: 9.65/10

Complutense University of Madrid

Degree in Physics. Speciality of Theoretical Physics.

2015–2020

Average: 8.79/10

Complutense University of Madrid

Degree in Mathematics. Speciality of Pure Mathematics.

2015–2020

Average: 8.65/10

PUBLICATIONS¹

- Fay, A.^{*}, **García-Redondo, I.^{*}**, Wang, Q.^{*}, Dubossarsky, H., and Monod, A.
The Shape of Adversarial Influence: Characterizing LLM Latent Spaces with Persistent Homology.
Accepted as an oral to The Fourteenth International Conference on Learning Representations (ICLR 2026).
<https://openreview.net/forum?id=v2Pg1vLLKT>.
- **García-Redondo, I.**, Monod, A., and Wang Q.
Confidence Bands for Multiparameter Persistence Landscapes.
Proceedings of the International Conference on Geometric Science of Information (GSI 2025),
Lecture Notes in Computer Science, 16034, Springer, 22–31.
https://doi.org/10.1007/978-3-032-03921-7_3. [Code]
- **García-Redondo, I.**, Landi, C., Percival, S., Skeja, A., Wang, B., and Zhou, L.
Finding the Cores of Higher Graphs Using Geometric and Topological Means: A Survey.
Accepted to Research in Computational Topology 3, Association for Women in Mathematics-Springer series.
ArXiv version.
- Tan, C.^{*}, **García-Redondo, I.^{*}**, Wang, Q.^{*}, Bronstein, M., and Monod, A.
On the Limitations of Fractal Dimension as a Measure of Generalization.
Advances in Neural Information Processing Systems, 37, 60309-60334. (NeurIPS 2024)
<https://openreview.net/forum?id=Y06GVPUrKN>.
- Wang, Q.^{*}, **García-Redondo, I.^{*}**, Faugère, P., Henselman-Petrusek, G. and Monod, A.
Stability for Inference with Persistent Homology Rank Functions.
Computer Graphics Forum, 43: e15142.
<https://doi.org/10.1111/cgf.15142>.
- **García-Redondo, I.**, Monod, A. and Song, A.
Fast topological signal identification and persistent cohomological cycle matching.
Journal of Applied and Computational Topology, 8(3), 695-726.
<https://doi.org/10.1007/s41468-024-00179-4>. [Code]

PREPRINTS

- Nordwald, N.^{*}, **García-Redondo, I.^{*}**, and Monod, A.
Density-Robust Spherical Coordinates from Persistent Cohomology
ArXiv version (26 pages).

¹For author lists in alphabetical order, all authors contributed equally, otherwise authors marked with * have an equal contribution.

- Tang, Y., Wang, Q., **García-Redondo, I***, and Monod, A.
Topological Signatures of Grokking
ArXiv version (19 pages).
- **García-Redondo, I.**, Landi, C., Percival, S., Skeja, A., Wang, B., and Zhou, L.
Effective Resistance in Simplicial Complexes as Bilinear Forms: Generalizations and Properties.
ArXiv version (39 pages).

TEACHING

- **Doctoral Course – Topological and Geometric Deep Learning: Theory, Methods, and Applications** (Complutense University of Madrid, Lecturer) Spring 2026
- **LSGNT Computing Course** (Imperial College, Lecturer, Senior TA) Winter 2024, 2023
- **MSc Project Co-supervision** (Imperial College, 4 projects) Summer 2024
- **MATH40004 – Calculus and Applications** (Imperial College, TA) Spring 2024, 2025
- **MATH40012 – Linear Algebra and Groups** (Imperial College, Senior TA) Winter 2023
- **MATH40001/40009 – Introduction to University Mathematics** (Imperial College, Senior TA, TA) Winter 2023, 2022
- **MATH40002 – Analysis 1** (Imperial College, Senior TA) Spring 2023
- **MATH50003 – Linear Algebra** (Imperial College, TA) Winter 2022
- **Basic Mathematics** (Complutense University of Madrid, Undergraduate TA) September–October 2018

RESEARCH GRANTS, AWARDS AND FELLOWSHIPS

- **G-Research Imperial PhD Prize** (6,000 GBP) G-Research
Award granted to PhD theses in mathematics and data science 2026
- **SNSF Scientific Exchange Grant** (36,740 CHF) Swiss National Science Foundation
Awarded to co-organize the Workshop of Topological Deep Learning 2026
- **ELBE Fellowship** (~200,000 EUR, declined) Max Planck Institute (MPI-CBG)
Fully-funded 3-year postdoctoral fellowship 2025
- **UKRI PhD Studentship** (~170,000 GBP) Engineering and Physical Sciences Research Council
Studentship covering tuition fees, a monthly stipend and a 13,000 GBP travel grant 2021–2025
- **Collaboration Grant** (2,000 EUR) University Complutense of Madrid
Grant awarded to carry out a research project under the supervision of María Pe Pereira 2020–2021
- **UCM Excellence Scholarship for Master’s Degrees** (1,800 EUR) University Complutense of Madrid
Scholarship covering 50% of a master’s degree tuition fees 2020–2021
- **Excellence Scholarship** (1,000 EUR) Community of Madrid
Award granted for academic merits 2019–2020 and 2016–2017

DISTINCTIONS AND HONORS

- **Best Student Paper Award** at The 7th international conference on Geometric Science of Information (2025)
- **Runner-up in the PhD Symposium Poster Competition** of the Department of Mathematics, Imperial College London.
Poster title: *On the Limitations of Fractal Dimension as a Measure of Generalization* (2024)
- Second prize at the local stage (Region of Murcia) of the **LI Spanish Mathematical Olympics** (2015)
- Second prize at the **XX Chemistry Olympics** of the Region of Murcia (2015)
- Third prize at the local stage (Region of Murcia) of the **XXV National Physics Olympics** (2015)
- Award of the Complutense University of Madrid for the **100 best access marks** (2015-2016)

SELECTED RESEARCH TALKS

- **Jornadas de Topología del dato** (Universidad de Sevilla) February 2026
Invited talk: “Effective Resistance in Simplicial Complexes: Generalizations and Properties”
- **Algebraic Topology: Methods, Science, and Computations** (ATMCS 12) January 2026
Contributed talk: “Effective Resistance in Simplicial Complexes: Generalizations and Properties”
- **EWM-EMS Summer School: Stability in topological data analysis** July 2025
Invited talk: “Stability for Inference with Persistent Homology Rank Functions”
- **Erlangen AI Hub: Mathematical Foundations of Intelligence Conference** June 2025
Invited talk: “On the Limitations of Fractal Dimension as a Measure of Generalization”

- **RSME's 7th Congress of Young Researchers** January 2025
Invited talk: "On the Limitations of Fractal Dimension as a Measure of Generalization"
- **Applied Topology Seminar (University of Oxford)** November 2024
Invited talk: "On the Limitations of Fractal Dimension as a Measure of Generalization"
- **ELBE Symposium (Center for Systems Biology Dresden)** November 2024
Invited talk: "Topological Approaches for Data Analysis and Machine Learning"
- **VI Encuentro Conjunto de la Real Sociedad Matemática Española y la Sociedad Matemática Mexicana** (Universidad Politécnica de Valencia) July 2024
Invited talk: "On the Limitations of Fractal Dimension as a Measure of Generalization"
- **Topological Data Analysis Seminar (Universidad Autónoma de Madrid)** March 2024
Invited talk: "Cycle Matching and Rank functions stability: two stories about Persistent Homology"
- **Applied Algebraic Topology Research Network Seminar** November 2023
Invited talk: "Computable Stability for Persistence Rank Function Machine Learning" (video)
- **Global Doctoral Academy - JADS Annual Symposium** November 2023
Invited talk: "Geometric and Topological Methods in Data Analysis and Machine Learning"
- **Algebraic Topology in Albany (ATiA) seminar** October 2023
Invited talk: "Computable Stability for Persistence Rank Function Machine Learning"
- **Spatial Analysis for Biological Imaging Workshop: SABI 2023** March 2023
Invited talk: "Fast Cohomological Cycle Matching"
- **Workshop of Young Researchers in Mathematics** September 2021
Invited talk: "Monodromy of isolated singularities in complex hypersurfaces"

OTHER PROFESSIONAL EXPERIENCE

- **Debating and public speaking trainer** (Colegio Mayor Juan XXIII Roncalli) 2020–2021
Activity based on giving courses in public speaking to the residents of the College, and preparing them to participate in debating tournaments.

SERVICE

- **Reviewer for NeurIPS and IEEE Transactions on Pattern Analysis and Machine Intelligence** 2026
- **Co-organizer of the Topological Deep Learning Workshop** at the University of Fribourg 2026
- **Project mentor at the London School of Geometry in Machine Learning** 2025
- **Co-organizer of a parallel session** (RSME's 7th Congress of Young Researchers) 2024
- **Reviewer for the Association for Women in Mathematics Springer Series** 2024
- **Co-organizer of the Imperial Junior Geometry Seminar** 2023–2024
- **Co-organizer of the Lunch Seminars** 2023–2023
- **Social Representative LSGNT** 2021–2025

EXTRA-CURRICULAR

- **Committee member of the Imperial College Choir** 2023–2025
Year 2024-2025: Publicity Officer.
Year 2023-2024: Tour manager. Organized a tour for 100 people to the North of Spain.
- **Student representative on the Faculty Board (UCM)** 2016–2018
- **University Debating**
Speaker at the World Universities' Debating Championship in Spanish (2018 and 2019); winner and best speaker across various tournaments in Spain.

LANGUAGES AND TECHNICAL SKILLS

Languages	Spanish native, English C2, French B1.
Programming languages	Matlab, Python, Julia.